A direct split-prime bound for the tenth discriminant

Working refinement: $q < 10^6$

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Abstract

Assuming the determinant construction and Lemmas 4.2, 5.1, and 6.2 of Banks, we obtain $q < 10^6$ for imaginary quadratic fields of class number one. This improves the preceding cutoff $5 \cdot 10^7$ by a factor of 50. The new ingredient is a direct upper bound on the weighted sum of split primes, obtained by counting primitive lattice points. It replaces both the theta-integral zero construction and the zero-dependent prime-bias estimate. Banks's determinant argument remains an external input. No Heegner method or small-conductor zero computation is used.

Status. This is a new working argument, with an exact rational certificate and finite checks of the lattice-counting formulas. The independent check of the preceding v2 argument does not cover this refinement. No new independent mathematical review is recorded here.

1 Result and inputs

Let $D < 0$ be a fundamental discriminant and put

$$q = |D|, \qquad K = \mathbb{Q}(\sqrt{D}), \qquad R = \sqrt{q}, \qquad L = \log q.$$

All logarithms are natural. Write χ_D for the associated quadratic character, so $\chi_D(p) = 1, -1, 0$ according as a rational prime splits, is inert, or ramifies in K.

Theorem 1. *Assume Banks's determinant construction and Lemmas 4.2, 5.1, and 6.2 in [1], including the interpolation results used in that construction. If $h(K) = 1$, then*

$$q < 1\,000\,000.$$

We also use the classical explicit prime estimates of Rosser–Schoenfeld [2, Theorems 6 and 9]. In particular, for $X > H \geq 319$,

$$\sum_{H < p \leq X} \frac{\log p}{p} > \log X - \log H - \frac{1}{\log H}, \qquad \vartheta(X) := \sum_{p \leq X} \log p < 2X. \tag{1}$$

The first follows by subtracting the two bounds in their Theorem 6; the constant term cancels. The second is weaker than their Theorem 9.

Unlike the previous argument [3], this proof uses neither Banks's Theorem 1.1 nor his Lemma 2.1 or Corollary 6.3. It uses the prime-power divisibility in Lemma 6.2 directly. It does not independently establish the cited determinant construction.

2 Counting nonrational integers

Choose the usual basis $\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega$, where $\operatorname{Re}\omega = u \in \{0, 1/2\}$ and $\operatorname{Im}\omega = R/2$. Thus

$$Q(a,b) := |a + b\omega|^2 = (a + ub)^2 + \frac{q}{4}b^2.$$

For $x \geq 0$, define

$$A(x) = \#\{(a,b) \in \mathbb{Z} \times \mathbb{Z}_{\geq 1} : Q(a,b) \leq x\}.$$

The restriction $b > 0$ counts one half of the nonrational points.

Lemma 2. *For every $x \geq 0$,*

$$\frac{\pi x}{R} - 2\sqrt{x} - \frac{2\sqrt{x}}{R} \leq A(x) \leq \frac{\pi x}{R} + \frac{2\sqrt{x}}{R}.$$

Proof. Put $B = 2\sqrt{x}/R$ and $f(t) = 2\sqrt{x - qt^2/4}$ on $[0, B]$. For each integer $1 \leq b \leq B$, the allowed real values of a form a closed interval of length $f(b)$. Its integer count lies between $f(b) - 1$ and $f(b) + 1$. Since f is decreasing,

$$\int_0^B f(t)\,dt - f(0) \leq \sum_{b=1}^{\lfloor B \rfloor} f(b) \leq \int_0^B f(t)\,dt.$$

The integral is $\pi x/R$, and there are at most B intervals. This gives both bounds, including when $B < 1$. □

Only primitive pairs can have prime norm. A small sieve already captures enough of this restriction. Set

$$P = 2 \cdot 3 \cdot 5 \cdot 7 = 210, \qquad A_P(x) = \#\{(a,b) : b \geq 1,\ Q(a,b) \leq x,\ \gcd(a,b,P) = 1\}.$$

Lemma 3. *For every $x \geq 0$,*

$$A_P(x) \leq \frac{\pi\rho}{R}x + B_R\sqrt{x}, \qquad \rho = \frac{768}{1225}, \quad B_R = \frac{88}{35} + \frac{192}{35R}.$$

Proof. Homogeneity of Q and inclusion–exclusion give the exact identity

$$A_P(x) = \sum_{d \mid P} \mu(d) A(x/d^2).$$

For positive coefficients use the upper bound of Lemma 2; for negative coefficients use its lower bound. The area coefficient is

$$\sum_{d \mid P} \frac{\mu(d)}{d^2} = \prod_{p \mid P}(1 - p^{-2}) = \rho.$$

Write

$$T = \sum_{d \mid P} \frac{|\mu(d)|}{d} = \frac{96}{35}, \qquad U = \sum_{d \mid P} \frac{\mu(d)}{d} = \frac{8}{35}.$$

The boundary coefficient is $2\sum_{\mu(d)=-1} d^{-1} + 2T/R = T - U + 2T/R = B_R$. □

3 Direct bounds on split and ramified primes

Assume $h(K) = 1$ and $q \geq 10^6$, and put $a_0 = q/4$. If p splits, its two prime ideals of norm p are principal. As the only units are ± 1, there are exactly four generators of norm p. They give two distinct pairs with $b > 0$. They are nonrational, and each satisfies $\gcd(a, b, P) = 1$, since $d \mid a, b$ would force $d^2 \mid Q(a, b) = p$. Consequently,

$$S_+(X) := \sum_{\substack{p \leq X \\ \chi_D(p)=1}} \frac{\log p}{p} \leq \frac{1}{2} \sum_{\substack{b \geq 1,\ Q(a,b) \leq X \\ \gcd(a,b,P)=1}} \frac{\log Q(a, b)}{Q(a, b)}. \tag{2}$$

All norms in this sum are at least a_0. For $X \geq a_0$, partial summation, followed by Lemma 3, therefore gives

$$\begin{aligned} S_+(X) &\leq \frac{1}{2} \left\{ A_P(X) \frac{\log X}{X} + \int_{a_0}^{X} A_P(t) \frac{\log t - 1}{t^2}\, dt \right\} \\ &\leq \frac{\pi\rho}{4R} \big((\log X)^2 - (\log a_0)^2 + 2 \log a_0 \big) + \frac{B_R(\log a_0 + 1)}{\sqrt{a_0}}. \end{aligned} \tag{3}$$

In the last line we dropped the nonpositive boundary term $-B_R(\log X + 2)/(2\sqrt{X})$. Here $\log a_0 > 2$, and $\log a_0 + 1 < L$ because $\log 4 > 1$. Also, for $R \geq 1000$,

$$\frac{\pi\rho}{4} < \frac{493}{1000}, \qquad B_R < \frac{63}{25}.$$

Thus the required direct split-prime bound is

$$S_+(X) < \frac{493}{1000R} (\log X)^2 + \frac{126L}{25R} \qquad (X \geq q/4). \tag{4}$$

A ramified prime likewise has a principal prime ideal of norm p. Its generator is nonrational, so $p \geq q/4$. Hence

$$S_0(X) := \sum_{\substack{p \leq X \\ p \mid q}} \frac{\log p}{p} \leq \frac{4}{q} \sum_{p \mid q} \log p \leq \frac{4L}{q}. \tag{5}$$

Neither this estimate nor the split-prime count requires a separate reduction to prime discriminants.

Combining (1), (4), and (5), for $X > H \geq 319$ and $X \geq q/4$ we obtain

$$\sum_{\substack{H < p \leq X \\ \chi_D(p)=-1}} \frac{\log p}{p} > \log X - \log H - \frac{1}{\log H} - \frac{493}{1000R} (\log X)^2 - \left(\frac{126}{25R} + \frac{4}{q} \right) L. \tag{6}$$

4 The determinant comparison

Take

$$H = 10^9, \qquad N = q^{13}, \qquad M = N^{4/3}, \qquad z = \frac{52}{3}, \qquad \log M = zL.$$

For $q \geq 10^6$, Banks's hypotheses $q \geq 9$, $1 \leq H \leq N$, and $H^{2/3} N^{4/3} \geq 2 \cdot 97^2$ hold, as do $M > H$ and $M \geq q/4$. Banks's determinant construction in $E = K(\sqrt{2})$ gives $\Delta \neq 0$, the nonzero integer $J = \mathrm{N}_{E/\mathbb{Q}}(\Delta)$, and degree sums S_1, S_{23}. Lemma 4.2 gives

$$\frac{S_{23}}{S_1} \leq \frac{768 \cdot 97^2}{10^9} < r := \frac{73}{10000}, \qquad \frac{N^4 M}{S_1} \leq \frac{4 \cdot 97^2}{10^6} < c := \frac{19}{500}.$$

Lemma 5.1 then implies

$$\frac{\log|J|}{4S_1} \le (1+r)\left(\log N + \frac{L}{2} + \log 8\right) + \frac{2c\log N}{M}. \tag{7}$$

For each prime $p > H$ with $\chi_D(p) = -1$, Lemma 6.2 supplies $p^{4E_p} \mid J$, where $E_p = \sum_{\alpha\in\mathcal{A}} \lfloor \alpha_1/p \rfloor \ge S_1/p - N^4$. Multiplying the divisibilities for $p \le M$, and using $\vartheta(M) < 2M$, gives directly

$$\frac{\log|J|}{4S_1} \ge \sum_{\substack{H<p\le M \\ \chi_D(p)=-1}} \frac{\log p}{p} - 2c. \tag{8}$$

This step uses no zero-dependent corollary.

Compare (7) with (8) and insert (6) at $X = M$. Since $\log N = \frac{3}{4}\log M$, division by L gives the necessary inequality

$$\left(\frac{1}{4} - \frac{3r}{4}\right) z \le \frac{493}{1000} z^2 \frac{L}{R} + \frac{1+r}{2} + \frac{126}{25R} + \frac{4}{q} + \frac{\log H + (1+r)\log 8 + 1/\log H + 2c}{L} + \frac{26c}{M}. \tag{9}$$

5 A uniform rational contradiction

The function $(\log q)/\sqrt{q}$ decreases for $q > e^2$. Using $q \ge 10^6$, we have the uniform estimates

$$\frac{L}{R} < \frac{13.816}{1000}, \qquad L > 13.815, \qquad R \ge 1000, \qquad \frac{\log H}{L} \le \frac{3}{2},$$
$$\log 8 < 2.08, \qquad \frac{1}{\log H} < 0.049, \qquad \frac{26c}{M} < 10^{-6}.$$

The last inequality follows already from $M \ge 10^6$ and $26c = 0.988 < 1$. The logarithmic estimates are certified by rational Taylor bounds for the exponential function in the accompanying program.

The left side of (9) equals

$$\left(\frac{1}{4} - \frac{3}{4}\frac{73}{10000}\right)\frac{52}{3} = \frac{127153}{30000} > 4.238.$$

Its right side is strictly less than the rational number

$$\mathcal{R} = \frac{493}{1000}\left(\frac{52}{3}\right)^2 \frac{13.816}{1000} + \frac{1.0073}{2} + \frac{3}{2} + \frac{1.0073(2.08) + 0.049 + 0.076}{13.815} + \frac{5.04}{1000} + \frac{4}{10^6} + \frac{1}{10^6} < 4.216.$$

Every terminating decimal here denotes its exact rational value. The gap is greater than 0.022. This contradicts (9), proving Theorem 1.

6 Relation to earlier versions and verification

The preceding v2 bound was $q < 5 \cdot 10^7$. It combined a theta-integral zero with a zero-free estimate extracted from Banks. The present proof instead obtains the necessary inert-prime mass directly from class number one. Its small sieve removes lattice points whose two coordinates share a factor among $2, 3, 5, 7$; it is not an exhaustive search through discriminants.

The earlier finite-discriminant sieve covers $q < 10^{10}$, so its unchanged certificates already cover the remaining interval below 10^6. They leave $3, 4, 7, 8, 11, 19, 43, 67, 163$. That finite classification is separate from the analytic cutoff proved here.

All 26 exact arithmetic checks and 22 finite sanity cases pass. To reproduce the exact rational certificate, run

```
python3 verify_direct_prime.py
```

Add `-sanity` to enumerate lattice points in finite examples, compare inclusion–exclusion with direct counting, and check that each split prime has two generators with positive imaginary part in the seven known fields of class number one with $q > 4$.

The finite checks do not prove the analytic lemmas or audit Banks's interpolation argument. Codex generated the new derivation, this note, and the checker. The preceding v2 cross-check remains scoped to v2.

References

[1] W. D. Banks, manuscript arXiv:2610.15483v1, especially the determinant construction in Sections 3–4 and Lemmas 4.2, 5.1, and 6.2.

[2] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94, Theorems 6 and 9. doi:10.1215/ijm/1255631807.

[3] *An explicit tenth-discriminant bound, version 2*, fixed release v0.2.0, 20 October 2026.